

Requirement Analysis Document

Project Name: Quiz App

Course Software Development 1

Programming Language: Java

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1. Introduction

1.1 Project Overview

Quiz App is a Java-based educational software application that allows users to answer multiple-choice questions and evaluate their knowledge. The application will present questions, calculate scores, and display results at the end of the quiz.

1.2 Purpose

The purpose of this project is to develop an interactive quiz application that helps users test their knowledge while demonstrating software development and object-oriented programming concepts.

1.3 Objectives

- Develop an interactive quiz application.
 - Allow users to answer multiple-choice questions.
 - Calculate and display scores.
 - Provide immediate feedback.
 - Apply software development principles and OOP concepts.
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2. Functional Requirements

FR-1: Start Quiz

The system shall allow users to start a new quiz.

FR-2: Display Questions

The system shall display quiz questions with multiple-choice options.

FR-3: Answer Selection

The system shall allow users to select an answer for each question.

FR-4: Next Question

The system shall allow users to move to the next question.

FR-5: Score Calculation

The system shall calculate the user's score based on correct answers.

FR-6: Result Display

The system shall display the final result after quiz completion.

FR-7: Timer

The system shall display the quiz duration or countdown timer.

FR-8: Restart Quiz

The system shall allow users to restart the quiz.

FR-9: Feedback System

The system shall provide feedback indicating whether an answer is correct or incorrect.

FR-10: About Section

The system shall provide information about the application and developer.

3. Non-Functional Requirements

NFR-1: Usability

The software should provide a simple and user-friendly interface.

NFR-2: Performance

The application should respond quickly to user actions.

NFR-3: Reliability

The application should run without crashes during normal operation.

NFR-4: Maintainability

The source code should be organized using object-oriented programming principles.

NFR-5: Portability

The application should run on any system supporting Java Runtime Environment (JRE).

4. System Requirements

Hardware Requirements

- Processor: Dual-Core Processor or Higher
- RAM: 4 GB or Higher
- Storage: 100 MB Free Space

Software Requirements

- Java Development Kit (JDK 17 or later)
 - Java Swing
 - Git
 - GitHub
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5. Assumptions and Constraints

Assumptions

- Users can read and understand quiz questions.
- Java Runtime Environment is installed on the target system.

Constraints

- The application will be developed using Java.
 - The project duration is limited to 10 weeks.
 - Development will follow Software Development course guidelines.
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6. Expected Outcome

The final software will provide an interactive quiz experience where users can answer questions, receive feedback, track their scores, and view final results. The application will include a graphical user interface, timer functionality, score calculation, and quiz management features.

Conclusion

Quiz App is designed to demonstrate software development practices, graphical user interface design, object-oriented programming, and user interaction. The project will provide an educational and engaging experience while fulfilling the requirements of the Software Development course.